DO NOW · AO1 REVISION

Mass number, atomic number and isotopes

Atomic structure
Paper 1 · Combined and Separate
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State the overall charge of an atom.**	Zero	☐
2	**Explain why atoms have an overall neutral charge.**	The number of protons equals the number of electrons.	☐
3	**State what is the same about all atoms of the same element.**	The number of protons	☐
4	**State what the atomic number of an atom tells you.**	The number of protons	☐
5 ★	**State what the mass number of an atom tells you.**	The total number of protons and neutrons	☐
6 ★	**Define isotopes.**	Atoms of the same element with different numbers of neutrons.	☐
7	**Which differs between two isotopes of an element: mass number or atomic number?**	The mass number	☐
8	**Describe how ions form from atoms.**	Atoms lose or gain electrons.	☐
9	**An atom of carbon is written $^{14}_{6}C$. State what the numbers 14 and 6 represent.**	14: the mass number (protons + neutrons). 6: the atomic number (protons).	☐
10 ★	**State how to calculate the number of neutrons in an atom.**	Number of neutrons = mass number − atomic number	☐
11 ★	**State the type of ion formed when an atom loses one or more electrons.**	A positive ion	☐

 · © 2026 PanPhy Labs. All rights reserved.